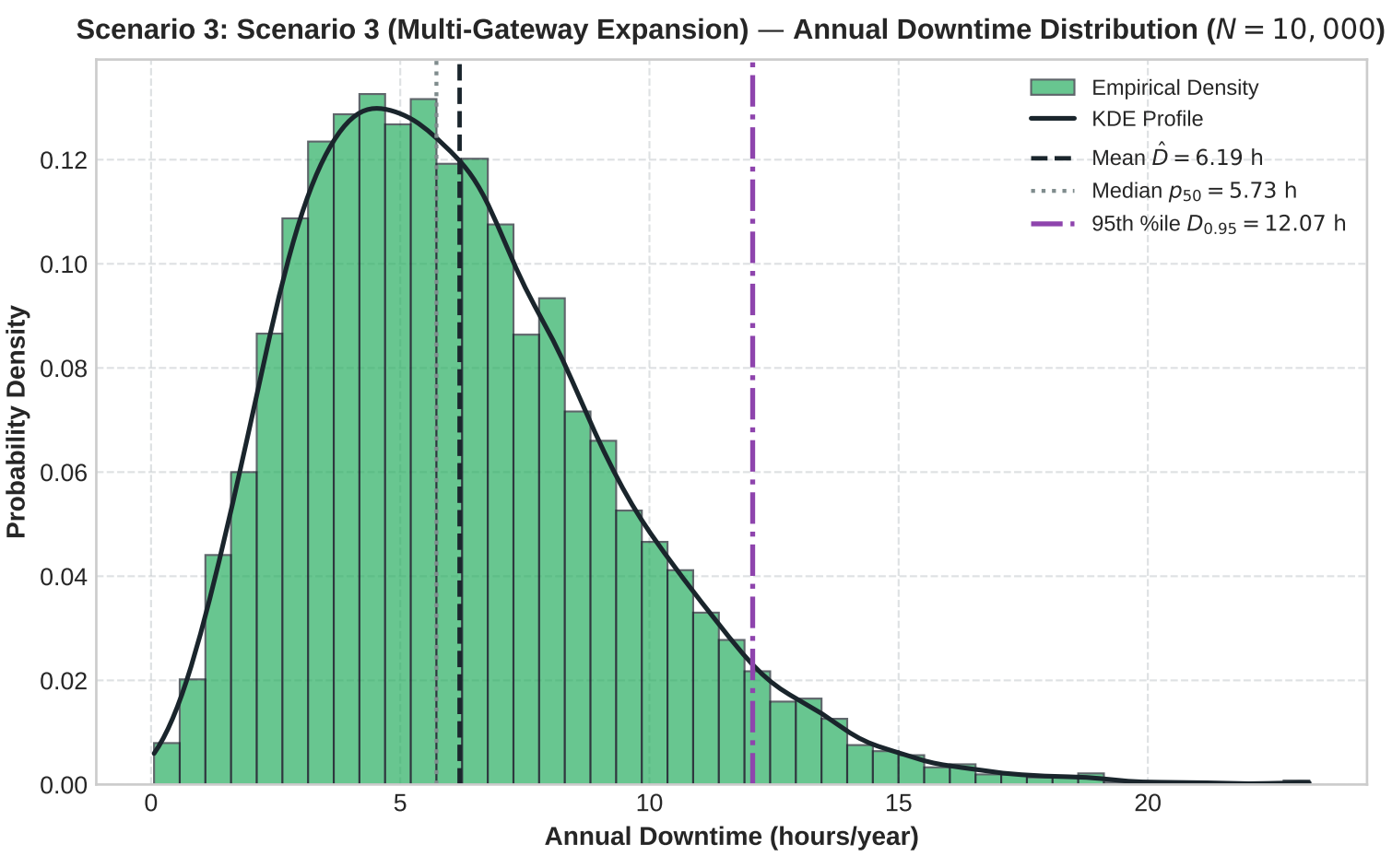
Scenario 3: Scenario 3 (Multi-Gateway Expansion) — Annual Downtime Distribution ($N = 10,000$)
Empirical Density
KDE Profile
Mean $\hat{D} = 6.19$ h
Median $p_{50} = 5.73$ h
95th %ile $D_{0.95} = 12.07$ h
Probability Density
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0
5
10
15
20
Annual Downtime (hours/year)